COMPUTER NETWORK.py - C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py (3.13.3)

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```python
import random

def generate_star_topology(n, switch_name="Main_Switch"):
    print("="*40)
    print(f"       STAR TOPOLOGY DIAGRAM")
    print("="*40)
    print(f"              [{switch_name}]")

    for i in range(1, n + 1):
        host_label = f"Host_{i}"
        # Generating a random length between 10m and 120m for realism
        cable_length = random.uniform(10.0, 120.0)
        print(f"                     |")
        print(f"                     | (Cat6 UTP: {cable_length:.2f} m)")
        print(f"                     |")
        print(f"                [{host_label}]")
        print()

# Example usage:
generate_star_topology(3, "Central_Switch")
```

Ln: 20 Col: 43

IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v
.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWOR
K.py =========
========================================
       STAR TOPOLOGY DIAGRAM
========================================
              [Central_Switch]
                   |
                   | (Cat6 UTP: 65.80 m)
                   |
              [Host_1]

                   |
                   | (Cat6 UTP: 92.38 m)
                   |
              [Host_2]

                   |
                   | (Cat6 UTP: 96.28 m)
                   |
              [Host_3]

>>>
```

Ln: 24 Col: 0

COMPUTER NETWORK.py - C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py (3.13.3)

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```python
def validate_segment(length_m):
    """
    Validates if the UTP segment is within IEEE 802.3 limits (<= 100m).
    """
    if length_m <= 100:
        return True
    else:
        print(f"Error: Segment length {length_m}m exceeds the IEEE 802.3 maximum limit of 100m.")
        return False

# --- Test Cases ---
print("Test Case 1 (45m):", validate_segment(45))   # Expected: True
print("Test Case 2 (100m):", validate_segment(100)) # Expected: True
print("Test Case 3 (125m):", validate_segment(125)) # Expected: False (with error message)
```

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py ==========
Test Case 1 (45m): True
Test Case 2 (100m): True
Error: Segment length 125m exceeds the IEEE 802.3 maximum limit of 100m.
Test Case 3 (125m): False
>>>
```

Ln: 9 Col: 0

Ln: 15 Col: 0

COMPUTER NETWORK.py - C:\Users\mr.dounut\Study\PYTHON\COMPUTER NETWORK.py (3.13.3)

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```python
import random

def simulate_csma_cd(attempts=5):
    hosts = ["Host_A", "Host_B", "Host_C", "Host_D"]
    log_filename = "csma_log.txt"

    with open(log_filename, "w") as log_file:
        for attempt in range(1, attempts + 1):
            # Randomly simulate 1 to 4 hosts trying to transmit at the exact same time
            transmitting_hosts = random.sample(hosts, random.randint(1, len(hosts)))

            if len(transmitting_hosts) > 1:
                # Collision occurs if more than 1 host transmits
                backoff = random.uniform(0.1, 5.0) # Random back-off in milliseconds
                log_msg = f"Event {attempt}: COLLISION DETECTED between {', '.join(transmitting_hosts)}. Back-off interval applied: {backoff:.2f} ms.\n"
                log_file.write(log_msg)
                print(log_msg.strip())
            else:
                # Successful transmission
                log_msg = f"Event {attempt}: SUCCESS. {transmitting_hosts[0]} transmitted successfully.\n"
                log_file.write(log_msg)
                print(log_msg.strip())

    print(f"\nSimulation complete. Results logged to '{log_filename}'.")

# Example usage:
simulate_csma_cd(10)
```

IDLE Shell 3.13.3

File Edit Shell Debug Options Window Help

```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

========= RESTART: C:\Users\mr.dounut\Study\PYTHON\COMPUTER NETWORK.py =========
Event 1: COLLISION DETECTED between Host_D, Host_C, Host_B, Host_A. Back-off interval applied: 4.44 ms.
Event 2: COLLISION DETECTED between Host_C, Host_B. Back-off interval applied: 0.93 ms.
Event 3: COLLISION DETECTED between Host_A, Host_D, Host_C, Host_B. Back-off interval applied: 3.73 ms.
Event 4: COLLISION DETECTED between Host_B, Host_D, Host_A, Host_C. Back-off interval applied: 4.21 ms.
Event 5: SUCCESS. Host_A transmitted successfully.
Event 6: COLLISION DETECTED between Host_B, Host_D, Host_A. Back-off interval applied: 0.82 ms.
Event 7: COLLISION DETECTED between Host_B, Host_C. Back-off interval applied: 2.94 ms.
Event 8: SUCCESS. Host_C transmitted successfully.
Event 9: COLLISION DETECTED between Host_B, Host_A, Host_D. Back-off interval applied: 2.62 ms.
Event 10: COLLISION DETECTED between Host_A, Host_C, Host_B, Host_D. Back-off interval applied: 4.81 ms.

Simulation complete. Results logged to 'csma_log.txt'.
>>>
```

Ln: 17 Col: 0

Ln: 27 Col: 0

COMPUTER NETWORK.py - C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py (3.13.3)

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```python
import socket
import threading
import time

# --- SWITCH (Server) ---
def star_switch_server(host='127.0.0.1', port=65432, messages_to_expect=2):
    mac_table = {}
    with socket.socket(socket.AF_INET, socket.SOCK_DGRAM) as sock:
        sock.bind((host, port))
        print("Switch: Powered on and listening...")

        for _ in range(messages_to_expect):
            data, addr = sock.recvfrom(1024)
            frame = data.decode().split('|') # Frame format: SourceMAC|DestMAC|Payload

            if len(frame) == 3:
                src_mac, dst_mac, payload = frame

                # Update MAC Table (Learning)
                mac_table[src_mac] = addr

                print(f"\n[Switch] Frame received from {src_mac}.")
                print(f"[Switch] MAC Address Table Updated: {mac_table}")
                print(f"[Switch] Forwarding message '{payload}' to {dst_mac}...")

# --- HOST (Client) ---
def host_client(src_mac, dst_mac, message, switch_ip='127.0.0.1', switch_port=65432):
    with socket.socket(socket.AF_INET, socket.SOCK_DGRAM) as sock:
        frame = f"{src_mac}|{dst_mac}|{message}"
        sock.sendto(frame.encode(), (switch_ip, switch_port))
        print(f"[Host {src_mac}] Transmitted frame to {dst_mac}.")

# --- Execution ---
if __name__ == "__main__":
    # Start the switch server in a separate thread
    switch_thread = threading.Thread(target=star_switch_server)
    switch_thread.start()

    time.sleep(1) # Wait for the switch to boot up

    # Host 1 sends a message to Host 2
    host_client("MAC:AA:11", "MAC:BB:22", "Hello from Host 1!")
    time.sleep(1)

    # Host 2 replies to Host 1
    host_client("MAC:BB:22", "MAC:AA:11", "Message received, Host 1!")

    switch_thread.join()
```

IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py =========
Switch: Powered on and listening...
[Host MAC:AA:11] Transmitted frame to MAC:BB:22.
[Switch] Frame received from MAC:AA:11.

[Switch] MAC Address Table Updated: {'MAC:AA:11': ('127.0.0.1', 49392)}
[Switch] Forwarding message 'Hello from Host 1!' to MAC:BB:22...

[Switch] Frame received from MAC:BB:22.[Host MAC:BB:22] Transmitted frame to MAC
:AA:11.

[Switch] MAC Address Table Updated: {'MAC:AA:11': ('127.0.0.1', 49392), 'MAC:BB:
22': ('127.0.0.1', 49393)}
[Switch] Forwarding message 'Message received, Host 1!' to MAC:AA:11...
>>>
```

Ln: 16 Col: 0

Ln: 48 Col: 24

COMPUTER NETWORK.py - C:/Users/mr.dounut/Study/PYTHON/COMPUTER NETWORK.py (3.13.3)

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```python
class StarSwitch:
    def __init__(self):
        self.ports = []

    def add_port(self, host_id):
        if host_id not in self.ports:
            self.ports.append(host_id)
            print(f"Success: Port added for {host_id}")
        else:
            print(f"Notice: {host_id} is already connected.")

    def remove_port(self, host_id):
        if host_id in self.ports:
            self.ports.remove(host_id)
            print(f"Success: Port removed for {host_id}")
        else:
            print(f"Error: {host_id} not found on any port.")

    def broadcast(self, frame, source_id):
        print(f"\n--- Initiating Broadcast from {source_id} ---")
        print(f"Payload: '{frame}'")

        if source_id not in self.ports:
            print(f"Error: Source {source_id} is not connected to the switch.")
            return

        for port in self.ports:
            if port != source_id:
                # Log each delivery except to the sender
                print(f"Log: Frame successfully delivered to {port}")

# --- Demonstration ---
switch = StarSwitch()
switch.add_port("Host_1")
switch.add_port("Host_2")
switch.add_port("Host_3")

switch.broadcast("ARP Request: Who has 192.168.1.5?", "Host_1")

switch.remove_port("Host_2")
switch.broadcast("Server shutting down", "Host_3")
```

Ln: 41 Col: 0

IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025,
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Enter "help" below or click "Help" above for more
information.
>>>
========= RESTART: C:/Users/mr.dounut/Study/PYTHON
/COMPUTER NETWORK.py =========
Success: Port added for Host_1
Success: Port added for Host_2
Success: Port added for Host_3

--- Initiating Broadcast from Host_1 ---
Payload: 'ARP Request: Who has 192.168.1.5?'
Log: Frame successfully delivered to Host_2
Log: Frame successfully delivered to Host_3
Success: Port removed for Host_2

--- Initiating Broadcast from Host_3 ---
Payload: 'Server shutting down'
Log: Frame successfully delivered to Host_1
>>>
```

Ln: 18 Col: 0